

The standing-wave bricks: certifier (a)’s instrument set, with the weak-side pipeline closed at every rank

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Abstract

Five bricks for certifier (a) of the rank-uniform registration program — the derivation of each rung’s standing wave from the carrier-lattice Poisson identity. (1) The μ_4 conjugation clock of the dual trace channel: conjugation is frequency reversal, so every even-rank dual term is real and every odd-rank dual term is purely imaginary, at every frequency. (2) The arithmetic unfolding of the compiled rank-uniform lattice FE-pair: the completed transform of the rank- r matrix bank is $\Lambda_r(s) = \pi^{-s}\Gamma(s) \sum_{n \geq 1} r_{(r+1)^2}(n) n^{-s}$ on the half-plane — the completed Epstein zeta of the carrier lattice, every rank, with its functional equation already compiled. (3) The weighted strand transformation law: the degree-one harmonically modulated strands exchange at shifted weight $3/2$ — the two-kernel, nontrivial-root shape that even-rank rungs require, imported from the compiled odd Hurwitz pair. (4) **The standing wave of the lattice family:** the completed transform is real on the central axis $\operatorname{Re} s = (r+1)^2/4$ at every rank — gate 2’s exact shape, derived from global Schwarz reality and the self-dual functional equation. The weak-side pipeline (geometric Poisson identity $\Rightarrow$ completed FE $\Rightarrow$ standing wave) is thereby executed end-to-end on a genuine arithmetic family, uniformly in the rank. The remaining wall of certifier (a) is located exactly — the amplitude exchange, one condition per rung, between two compiled endpoints — and is typed and *executed at* GL(1): the Dirichlet rung inhabits the exchange in full, gate, continuation, standing wave, and functional equation all compiled.

1 The μ_4 conjugation clock

The carrier-native dual channel at rank r (`RankLadderParity.lean`) is

$$I_r(\xi) = \int_{-1}^1 \Phi(x) U_r(x) e^{-2\pi i x \xi c} dx,$$

with Φ the even orbital profile and U_r the Chebyshev Satake character. The compiled parity clock $I_r(-\xi) = (-1)^r I_r(\xi)$ now composes with the new conjugation clock (`dualIntegral.conj`: $\overline{I_r(\xi)} = I_r(-\xi)$, no hypotheses) into the μ_4 law $\overline{I_r(\xi)} = (-1)^r I_r(\xi)$: every even-rank dual term is *real* (`dualIntegral.real.even`) and every odd-rank dual term is *purely imaginary* (`dualIntegral.imaginary.odd`) — at every rank, frequency, scale, and even profile. The reality half of the axis condition is compiled on the dual channel.

*Machine-checked companions: five files in `RequestProject/`, named per section below, carrying 58 named declarations in total; every declaration at axiom footprint `{propext, Classical.choice, Quot.sound}` or a subset, no `sorry`, no `axiom`, zero `sorryAx`.

2 The arithmetic unfolding of the lattice bank

The rank- r matrix bank

$$\theta_r(t) = \sum_{A \in M_{r+1}(\mathbb{Z})} e^{-\pi t \|A\|^2}$$

carries a compiled completed FE at every rank (the lattice FE-pair, `CarrierLatticePair.lean`). `LatticeThetaUnfolding.lean` identifies its chart. The grouping identity `bankGauss_eq_repSum` regroups the lattice sum by norm fibers, with finite fibers and polynomial growth $r_N(n) \leq (2n+1)^N$ of the representation numbers, so the bank is the coefficient theta of r_N against the exponential clock, whose Mellin transform is $\pi^{-s}\Gamma(s)$. On the half-plane, at every rank (`matrixLattice_arithmetic_chart`):

$$\Lambda_r(s) = \pi^{-s} \Gamma(s) \sum_{n \geq 1} r_{(r+1)^2}(n) n^{-s}, \quad \Lambda_r\left(\frac{(r+1)^2}{2} - s\right) = \Lambda_r(s),$$

the second equation already compiled. This is the completed Epstein zeta of $\mathbb{Z}^{(r+1)^2}$, produced by the house lattice transformation law with its arithmetic readout attached — a genuine arithmetic family of completed functional equations, uniform in the rank, fiber-free.

3 The weighted strand

The unweighted strand transforms to itself at weight $1/2$ with root number 1. The degree-one harmonically modulated strands — the compiled odd Hurwitz pair — transform to *each other* at shifted weight $3/2$ (`WeightedStrandTransformation.lean`: `sinKernel_inv`, `oddKernel_inv`):

$$\sin_a(1/t) = t^{3/2} \text{odd}_a(t), \quad \text{odd}_a(1/t) = t^{3/2} \sin_a(t).$$

Two different kernels, exchange closing only after two reflections, weight shifted by the harmonic degree: the $d = 1$ rung of the Hecke modulation ladder $P \cdot \text{gauss} = i^{-d} P \cdot \text{gauss}$, and the exact geometric shape of nontrivial root structure. The $\Gamma_{\mathbb{R}}(s+1)$ -completed Mellin side and its functional equations are compiled upstream.

4 The standing wave of the lattice family

Theorem 1 (the standing wave; machine-checked). *At every rank r , the completed transform of the matrix bank satisfies, on the central axis $\text{Re } s = (r+1)^2/4$:*

$$\Lambda_r(s) = \overline{\Lambda_r(s)}$$

— gate 2's FE-on-axis shape with root number 1; equivalently the transform is real on the axis (`matrixLattice_axis_im_zero`), and its entire part carries the same standing wave. The proof composes global Schwarz reality $\Lambda_r(\bar{s}) = \overline{\Lambda_r(s)}$ (`matrixLattice_schwarz`: the Mellin transform of a real profile is Schwarz-real on the half-plane, globalized through the entire part by the identity theorem) with the self-dual completed functional equation (`matrixLattice_FE`; the pair is its own reflection, `matrixLatticePair_symm_eq`). No new analytic input.

The weak-side pipeline of certifier (a) — geometric Poisson identity $\Rightarrow$ completed FE $\Rightarrow$ standing wave — is thereby closed, end-to-end, on a genuine arithmetic family, at every rank simultaneously. Home: `LatticeStandingWave.lean`, theorem `matrixLattice_standingWave`, capstone `lattice_weak_rung`.

5 The exchange, executed at GL(1)

The wall is now a type (`AmplitudeExchange.lean`): an `ExchangedRung` is a cancellation-gated rung together with its standing wave, and `ExchangedRung.fe` runs the compiled reduction, so inhabiting the type at a profile *is* that profile's functional equation on the strip. The first full inhabitant is compiled:

Theorem 2 (the Dirichlet rung; machine-checked). *Let χ be a nontrivial Dirichlet character mod q . Then: (i) its partial sums are bounded by q — full windows vanish exactly — so χ inhabits the cancellation gate at weight 1 with exponent 0 (`chiGate`); (ii) the gate's Abel continuation is the entire Dirichlet L -function on $\text{Re } s > 0$ (`abelContinuation_eq_LFunction`, identity theorem from the convergence region); (iii) the conjugate amplitude is the inverse character (`conj_chi_apply`: unit values have modulus one), so the compiled Schwarz pairing exchanges χ and χ^{-1} ; (iv) for the conductor-completed chart $q^{s/2}\gamma(\chi, s)$ (`chiRungC`), if χ^{-1} is primitive with nonvanishing root number, the rung's **standing wave holds** (`chiRungC_standingWave`): on the axis $\bar{s} = 1 - s$, the Schwarz pairing and the compiled completed functional equation collide, the conductor powers recombine as $q^{(1-s)/2} q^{s-1/2} = q^{s/2}$, and the readout is ε -real with the constant $\varepsilon = W(\chi^{-1})^{-1}$. Hence `chiExchangedRung` inhabits the exchange, and `chi_fe` delivers the completed Dirichlet functional equation on the critical strip through the reduction.*

The pipeline gate $\Rightarrow$ continuation $\Rightarrow$ standing wave $\Rightarrow$ FE is thereby executed end-to-end on a genuinely cancellation-gated arithmetic amplitude — the GL(1) prototype of the amplitude exchange. The inputs consumed are compiled Mathlib theorems (the completed functional equation for primitive characters, itself machine-checked, plus one nonvanishing datum: the primitive Gauss sum), not citations.

6 Register

Remark 3 (register). *Proved (machine-checked). All five bricks, as named above — including the GL(1) exchange, whose only unconsumed datum is the nonvanishing of the primitive root number (with primitivity of χ^{-1} carried as a hypothesis).*

The obstruction, measured. *The rank-uniform reduction's cuspidal gate (`TransferGate`) demands sub-axis primitive cancellation $\|\sum_{k < n} a_k\| \leq Cn^\theta$ with $\theta < \kappa/2$. The lattice family cannot supply it: its partial sums grow at the full abscissa — the two poles of Λ_r , residues ± 1 , are exactly the measured non-cancellation of the DC mode. The weak side is therefore the lattice family's home, and the standing wave above is its gate-2 statement.*

The remaining wall, located. *Certifier (a)'s outstanding input is the amplitude exchange: replace the representation-number amplitude by a rung profile carrying the cancellation gate, while keeping the lattice transformation law. One condition per rung, now positioned between two compiled endpoints — the geometric Poisson identity with its standing wave on one side, the gate-and-reduction machinery (`RankUniformReduction`) on the other.*

Cited, not formalized. *The classical discharge of the exchanged rungs' functional equations is symmetric-power automorphy: Kim–Shahidi at $r = 3, 4$ [1, 2] and Newton–Thorne in general [3]; nothing here consumes them.*

Not claimed. *No rung's standing wave beyond the lattice family is claimed; the dead coupling routes recorded in the program ledger (strand modulation, kernel telescoping, Veronese folding, aligned bilateral summation) remain dead.*

Falsifier. *A rank at which the lattice family’s transform fails axis reality would refute the standing wave and, through it, the compiled Schwarz or FE bricks; a rung amplitude satisfying the cancellation gate whose completed readout fails the reduction’s FE would refute the gate capstone.*

References

- [1] H. Kim, F. Shahidi, *Functorial products for $\mathrm{GL}_2 \times \mathrm{GL}_3$ and the symmetric cube for GL_2* , Ann. of Math. (2) **155** (2002), 837–893.
- [2] H. Kim, *Functoriality for the exterior square of GL_4 and the symmetric fourth of GL_2* , J. Amer. Math. Soc. **16** (2003), 139–183.
- [3] J. Newton, J. A. Thorne, *Symmetric power functoriality for holomorphic modular forms, I–II*, Publ. Math. IHÉS **134** (2021), 1–116 and 117–152.